

Model Comparison for California Housing Price Prediction

Methodology, Results, and Conclusions

Overview

This report summarizes an end-to-end regression analysis conducted on the California Housing dataset. The objective was to predict median house values from demographic and geographic features, and to compare the performance of three regression algorithms: Linear Regression, Ridge Regression, and Decision Tree Regression.

1. Methodology

1.1 Dataset

The analysis used the California Housing dataset (fetched via scikit-learn's `fetch_california_housing` utility), consisting of 20,640 records and 8 numerical features: median income, house age, average rooms, average bedrooms, block population, average occupancy, latitude, and longitude. The target variable, `MedHouseVal`, represents the median house value for a district, expressed in units of \$100,000.

1.2 Data Preparation

- Train-test split:** The data was split into 80% training (16,512 rows) and 20% testing (4,128 rows) subsets, using a fixed `random_state` of 42 for reproducibility.
- Feature scaling:** A `StandardScaler` was fit on the training features and used to transform both the training and test sets, ensuring all features were standardized (zero mean, unit variance) before model training. The scaler was fit only on training data to avoid data leakage.

1.3 Models Trained

- Linear Regression:** A baseline ordinary least squares model with no regularization.
- Ridge Regression:** A linear model with L2 regularization ($\alpha = 1.0$), intended to reduce overfitting by penalizing large coefficients.
- Decision Tree Regressor:** A non-linear, tree-based model (`random_state = 42`) capable of capturing non-linear relationships and feature interactions.

1.4 Evaluation Metrics

Each model was evaluated on the held-out test set using two standard regression metrics: **RMSE (Root Mean Squared Error)**, which measures the average magnitude of prediction error in the same units as the target (lower is better), and **R² (coefficient of determination)**, which measures the proportion of variance in house values explained by the model (higher is better, max = 1.0).

2. Results

2.1 Performance Comparison

Model	RMSE	R ² Score
Decision Tree	0.7060	0.6196
Ridge Regression	0.7456	0.5758

Linear Regression	0.7456	0.5758
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The Decision Tree Regressor achieved the lowest RMSE (0.706) and highest R^2 (0.620), outperforming both linear models. Ridge Regression and Linear Regression produced nearly identical results, indicating that the modest L2 penalty ($\alpha = 1.0$) had little effect on this dataset, likely because multicollinearity or overfitting were not severe concerns for the plain linear model.

2.2 Actual vs. Predicted Values

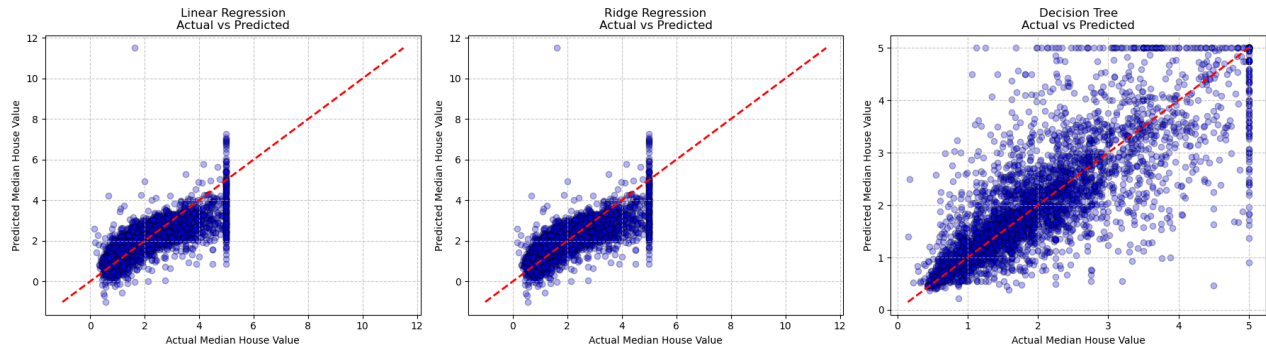


Figure 1: Scatter plots of actual vs. predicted median house values for each model, with the red dashed line representing perfect prediction ($y = x$). The Decision Tree's points cluster somewhat closer to the ideal line, though all three models show increasing spread and a ceiling effect near the upper end of the value range (a known artifact of the dataset, which caps house values at \$500,000).

2.3 Residual Analysis and Feature Importance

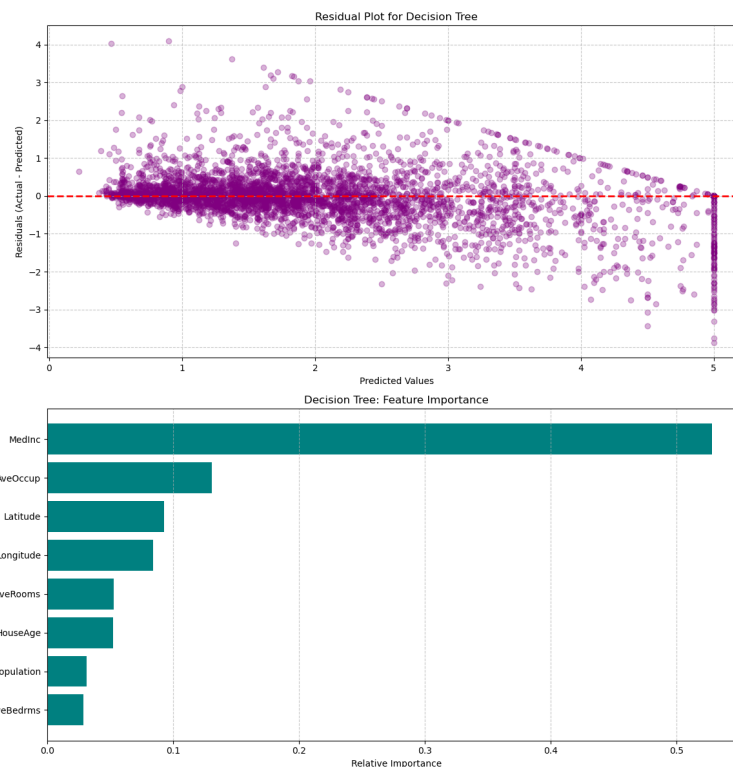


Figure 2 (top): Residual plot for the best-performing model (Decision Tree). Residuals are reasonably centered around zero but show a fan-shaped pattern of increasing variance at higher predicted values, suggesting mild heteroscedasticity. Figure 2 (bottom): Feature importances from the Decision Tree show that **median income (MedInc)** is by far the most influential predictor of house value, followed by geographic location (latitude and longitude), with average occupancy and population contributing comparatively little.

2.4 Model Persistence

The best-performing model (Decision Tree) was serialized using **joblib** and saved to disk as **best_california_housing_model.pkl**, making it available for reuse in downstream applications (e.g., a Streamlit interface) without retraining.

3. Conclusions and Recommendations

The Decision Tree Regressor outperformed both linear models on this dataset, suggesting that the relationship between the input features and median house value is not purely linear, and that the tree's ability to model interactions and thresholds (e.g., income brackets, geographic clusters) provides a modeling advantage. However, an R^2 of 0.62 still leaves roughly 38% of the variance in house prices unexplained, and single decision trees are prone to overfitting and instability on unseen data.

Suggested next steps:

- **Ensemble methods:** Random Forest or Gradient Boosting (e.g., XGBoost) would likely improve accuracy and generalization over a single decision tree by reducing variance.
- **Hyperparameter tuning:** Tuning tree depth/min-samples-leaf, or the Ridge alpha parameter, via cross-validation could improve results and reduce overfitting risk.
- **Feature engineering:** Combining latitude/longitude into derived features (e.g., distance to coast or major cities) or creating rooms-per-household ratios may boost predictive power.
- **Addressing the value cap:** House values capped at \$500,000 introduce a data artifact; treating capped records separately or using a model robust to censored data could improve accuracy at the high end.